

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular & Supplementary Winter Examination-2023**

**Course: B. Tech.**

**Semester :VII**

**Branch : Electronics & Allied Engg.**

**Subject Code & Name: BTHM705\_EX Engineering Economics and Financial Mathematics**

**Max Marks: 60**

**Date: 11-01-24**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                                                          | CO           | Marks     |             |      |          |      |               |      |                         |      |                |      |              |              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|-------------|------|----------|------|---------------|------|-------------------------|------|----------------|------|--------------|--------------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                              |              | <b>12</b> |             |      |          |      |               |      |                         |      |                |      |              |              |
| A) Explain in detail the different element of cost.                                                                                                                                                                                                                                                                                                      | CO1          | 6         |             |      |          |      |               |      |                         |      |                |      |              |              |
| B) Explain different types of Efficiency in Engineering Economics.                                                                                                                                                                                                                                                                                       | CO1          | 6         |             |      |          |      |               |      |                         |      |                |      |              |              |
| C) What is process planning and give steps for the same?                                                                                                                                                                                                                                                                                                 | CO1          | 6         |             |      |          |      |               |      |                         |      |                |      |              |              |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                               |              | <b>12</b> |             |      |          |      |               |      |                         |      |                |      |              |              |
| A) Mention a basic steps for value engineering.                                                                                                                                                                                                                                                                                                          | CO1          | 6         |             |      |          |      |               |      |                         |      |                |      |              |              |
| B) The Management of a company finds that while the cost of making a component part is Rs.10, the same is available in the market at Rs.9 with an assurance of continuous supply. Give a suggestion whether to make or buy this part. Give also your views in case the supplier reduces the price from Rs.9 to Rs.8. The cost information is as follows: | CO1          | 6         |             |      |          |      |               |      |                         |      |                |      |              |              |
| <table><tr><th>Particulars</th><th>R.S.</th></tr><tr><td>Material</td><td>3.50</td></tr><tr><td>Direct Labour</td><td>4.00</td></tr><tr><td>Other variable expenses</td><td>1.00</td></tr><tr><td>Fixed expenses</td><td>1.50</td></tr><tr><td><b>Total</b></td><td><b>10.00</b></td></tr></table>                                                       |              |           | Particulars | R.S. | Material | 3.50 | Direct Labour | 4.00 | Other variable expenses | 1.00 | Fixed expenses | 1.50 | <b>Total</b> | <b>10.00</b> |
| Particulars                                                                                                                                                                                                                                                                                                                                              | R.S.         |           |             |      |          |      |               |      |                         |      |                |      |              |              |
| Material                                                                                                                                                                                                                                                                                                                                                 | 3.50         |           |             |      |          |      |               |      |                         |      |                |      |              |              |
| Direct Labour                                                                                                                                                                                                                                                                                                                                            | 4.00         |           |             |      |          |      |               |      |                         |      |                |      |              |              |
| Other variable expenses                                                                                                                                                                                                                                                                                                                                  | 1.00         |           |             |      |          |      |               |      |                         |      |                |      |              |              |
| Fixed expenses                                                                                                                                                                                                                                                                                                                                           | 1.50         |           |             |      |          |      |               |      |                         |      |                |      |              |              |
| <b>Total</b>                                                                                                                                                                                                                                                                                                                                             | <b>10.00</b> |           |             |      |          |      |               |      |                         |      |                |      |              |              |
| C) A company has to replace a present facility after 15 years at an outlay of Rs.5,00,000. It plans to deposit an equal amount at the end of every year for the next 15 years at interest rate of 18% compounded annually. Find the equivalent amount that must be deposited at the end of every year for the next 15 years.                             | CO1          | 6         |             |      |          |      |               |      |                         |      |                |      |              |              |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                              |              | <b>12</b> |             |      |          |      |               |      |                         |      |                |      |              |              |
| A) Explain Revenue Dominated Cash flow diagram?                                                                                                                                                                                                                                                                                                          | CO1          | 6         |             |      |          |      |               |      |                         |      |                |      |              |              |

- B)** A Transport company has been looking for a new tyre for its truck and has located the following alternatives: CO1 **6**

| Brand | Tyre warranty (month) | Price per tyre (R.s.) |
|-------|-----------------------|-----------------------|
| A     | 12                    | 1,200                 |
| B     | 24                    | 1,800                 |
| C     | 36                    | 2,100                 |
| D     | 48                    | 2,700                 |

If the company feels that the warranty period is a good estimate of the tyre life and that a normal interest rate (compound annually) of 12% is appropriate, which tyre should it buy?

- C)** Alpha industry is planning to expand its production operation. It has identified three different Technology for meeting the goal. The initial outlay and annual revenues with respect to each of the technologies are summarized in table. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compound annually. CO1 **6**

|             | Initial outlay(Rs.) | Annual revenue (Rs.) | Life(years) |
|-------------|---------------------|----------------------|-------------|
| Technology1 | 12,00,000           | 4,00,000             | 10          |
| Technology2 | 20,00,000           | 5,00,000             | 10          |
| Technology3 | 18,00,000           | 6,00,000             | 10          |

**Q.4 Solve Any Two of the following.** **12**

- A)** Write down the cause of Equipment breakdown and Give disadvantages of Breakdown Maintenance? CO1 **6**
- B)** Write a short note on Preventive Maintenance? CO1 **6**
- C)** Two years ago, a machine was purchased at a cost of Rs. 2,00,000 to be useful for eight years. Its salvage value at the end of its life is Rs. 25,000. The annual maintenance cost is Rs. 25,000. The market value of the present machine is Rs. 1,20,000. CO1 **6**

Now, a new machine to cater to the need of the present machine is available at Rs. 1,50,000 to be useful for six years. Its annual maintenance cost is Rs. 14,000. The salvage value of the new machine is Rs. 20,000. Using an interest rate of 12%, find whether it is worth replacing the present machine with the new machine.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|
| <b>Q. 5 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     | <b>12</b> |
| <b>A)</b> Differentiate straight line method of depreciation and declining balance method of depreciation?                                                                                                                                                                                                                                                                                                                                                                                                                            | CO1 | <b>6</b>  |
| <b>B)</b> Two mutually exclusive projects are being considered for investment. Project A1 requires an initial outlay of Rs. 30, 00,000 with net receipts estimated as Rs. 9, 00,000 per year for the next 5 years. The initial outlay for the project A2 is Rs. 60, 00,000, and net receipts have been estimated at Rs. 15, 00,000 per year for the next seven years. There is no salvage value associated with either of the projects. Using the benefit cost ratio, which project would you select? Assume an interest rate of 10%. | CO1 | <b>6</b>  |
| <b>C)</b> Discuss different types of inflation?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CO1 | <b>6</b>  |

**\*\*\* End \*\*\***